



CT-Mono: Leveraging CNNs and Transformers for Self-Supervised Depth Estimation in Single-View Scenarios

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Abstract

Although self-supervised monocular depth estimation methods have made significant progress in computer vision, current real-time depth estimation methods based on visual features exhibit low accuracy in reconstructing fine structures in complex scenes. This paper introduces CT-Mono, a self-supervised monocular depth estimation framework that integrates convolutional neural networks and transformers. It enhances local representation through a multi-scale feature fusion mechanism and constructs a cross-region attention network to capture global contextual associations, thereby facilitating the accurate reconstruction of fine structures in complex scenes. First, leveraging the Depthwise Separable Dilated Convolution Block (DCB), multi-layer dilated convolution expands the receptive field, while the fusion of multi-scale pixel features enhances local details in key regions. Second, Enhanced Local–Global Features Fusion is employed alongside local convolution to capture fine-grained local image information, while the self-attention mechanism captures long-range dependencies. This synergistic modeling of local and global information significantly enhances feature representation. Finally, the reprojection loss is formulated by estimating the six-degree-of-freedom relative transformation between neighboring images using a lightweight encoder based on the pre-trained ResNet18, thereby optimizing depth prediction. Experiments on the KITTI dataset demonstrate that CT-Mono achieves an AbsRel of 0.102 and an RMSE of 4.430, significantly outperforming existing mainstream methods.

Keywords Monocular depth estimation · Self-supervision · CT-Mono · Multi-stage encoder

1 Introduction

Depth estimation is a fundamental problem in computer vision, demonstrating significant value and broad application prospects in various real-world scenarios. These applications include robot navigation and environment perception [1], medical image processing, autonomous driving [2, 3], 3D reconstruction [4], and augmented reality [5], among others. However, a key challenge in depth estimation is that a given 2D image may correspond to multiple possible 3D scenes, making monocular depth estimation inherently ill-posed. This arises because a single image typically lacks sufficient information to uniquely determine an object's structure in

3D space [6]. Earlier studies employed traditional methods [7–10] to infer depth by analyzing various cues in a single image, such as viewpoint information, texture features, and similarities to known scene structures. However, these methods heavily rely on hand-crafted features and geometric constraints, posing significant challenges and limitations in achieving accurate depth estimation from a single image.

In practical applications, while depth sensors [11] can provide relatively accurate depth information, their high cost often limits their feasibility in certain scenarios. Consequently, researchers have explored deep learning-based methods to efficiently estimate depth from monocular images. Depending on the need for real depth values, monocular depth estimation algorithms can be broadly categorized into three types: supervised, semi-supervised, and self-supervised. Supervised depth estimation methods train models using depth maps with real ground truth, allowing them to learn to accurately associate each image pixel with its corresponding depth value. The advantage of this approach is that it eliminates reliance on traditional depth sensors, enabling direct inference of per-pixel depth from image data. How-

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ever, this approach requires large-scale, high-quality ground truth depth data for training, motivating the development of self-supervised methods.

Self-supervised depth estimation relies heavily on the geometric relationships between consecutive frames. This method trains the model by analyzing camera motion and pixel changes across consecutive frames while minimizing image reconstruction loss. Specifically, the process transforms a frame using predicted depth and camera motion and then compares the transformed image with the original frame to minimize differences, enabling the network to learn depth-related features [12].

To enhance monocular depth estimation performance, researchers have explored various deep neural network (DNN)-based models [13–15], where DNNs employ fully convolutional architectures as feature extractors [16]. However, repeated spatial pooling layers can significantly reduce the spatial resolution of feature maps, which adversely affects depth estimation. Although high-resolution depth maps can be generated using multilayer deconvolution networks [17] or skip connections [18], these approaches often demand additional computational resources and increase network complexity. To efficiently extract depth-related features, convolutional neural networks (CNNs) are commonly employed as the backbone of deep models. CNN-based deep networks typically consist of an encoder for feature extraction and a decoder for depth prediction. While many studies have focused on decoder design [19, 20], recent research highlights the encoder's critical role in accurate depth estimation [21]. Popular encoders in encoder–decoder architectures include ResNet [22], DenseNet [23], and ResNeXt [24], which are designed to extract essential features. Given the absence of depth cues, efficient encoders must effectively capture long-range correlations (e.g., object distance relationships) and local continuity (e.g., within the same object) [6]. Thus, the primary bottleneck in existing depth estimation methods lies in the encoder, as convolution operations have a limited receptive field, making it difficult to model long-range dependencies accurately. Depth maps often exhibit unclear object boundaries, as shown in Fig. 1, making it challenging for traditional methods to recover fine scene details.

To address the limitations of existing methods, this paper proposes a multi-stage encoder architecture that integrates Convolutional Neural Networks (CNNs) and Transformers. This architecture effectively expands the receptive field and fully leverages global contextual information by combining the local feature extraction capabilities of CNNs with the global modeling capacity of Transformers. Additionally, cascading modules are introduced to fuse fine-grained details extracted at different stages with multi-scale features. This multi-scale network structure enables simultaneous attention to both global and local features, facilitating high-resolution detail recovery in depth maps.

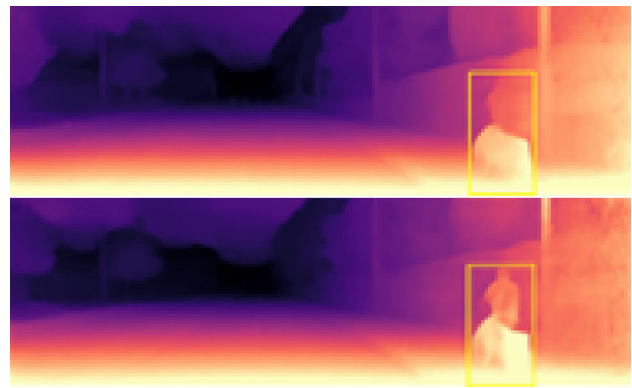


Fig. 1 Depth map (Comparison of object boundaries)

The contributions of this paper can be summarized in the following three aspects:

1. A novel self-supervised monocular depth estimation framework (CT-Mono): This paper presents CT-Mono, an innovative self-supervised learning framework that efficiently leverages visual geometric relationships and motion cues between adjacent frames. By optimizing geometric consistency and photometric consistency losses, CT-Mono achieves high accuracy and robustness in depth estimation, eliminating the reliance on expensive depth sensors or large-scale annotated datasets.

2. Depthwise Separable Dilated Convolution Block (DCB) module: To enhance feature extraction efficiency and effectiveness, this paper proposes DCB, which combines the strengths of Deeply Separable Convolution and Dilated Convolution. Deeply Separable Convolution efficiently captures local features, while Dilated Convolution expands the receptive field to incorporate broader contextual information. This fusion approach not only reduces model complexity but also enhances the network's ability to preserve fine-grained details and global structures, thereby improving depth estimation accuracy and robustness.

3. Enhanced Local–Global Features Fusion (ELGF) module: This paper introduces the ELGF module, which adaptively adjusts feature representations across different spatial scales using parallel local convolution and attention mechanisms. By improving structural perception at varying scales, the ELGF module significantly enhances depth estimation accuracy, particularly in complex scenes.

The rest of the paper is organized as follows. In Sect. 2, the related work is presented. The main components of the proposed method are described in Sect. 3. Section 4 contains reports on related experiments. Conclusions and future work are summarized in Sect. 5.

2 Related Works

In recent years, the field of Monocular Depth Estimation (MDE) has made significant progress, driven by the integration of deep learning and self-supervised learning paradigms. This section reviews important contributions and methods for supervised, self-supervised MDE and related network architecture improvements, focusing on technological developments that have significantly improved the accuracy of depth prediction for single images.

2.1 Supervised Depth Estimation

Monocular depth estimation aims to infer 3D depth information from a single 2D image. Supervised depth estimation methods enhance model accuracy by training with real depth labels, typically obtained from RGB-D cameras or LiDAR sensors. Eigen et al. [25] first introduced a multi-scale depth neural network that refines depth prediction by combining coarse and fine-grained depth maps. Later, Laina et al. [26] leveraged Residual Networks (ResNet) for depth prediction, proposing an upsampling module and inverse Huber loss, which significantly improved training performance. Additionally, Kendall et al. [27] introduced the L1 loss function to optimize the training process by minimizing the absolute difference between predicted and ground-truth depth maps, thereby enhancing model robustness and accuracy. They also explored an end-to-end learning framework that integrates geometric and contextual information, improving performance in complex scenes. Fu et al. [28] redefined depth regression as a classification problem with discrete depth values, leading to substantial improvements in accuracy and convergence speed. Building on this, the AdaBins method [29] was developed, which further enhances flexibility and precision by adaptively adjusting depth intervals for each input image.

2.2 Self-Supervised Depth Estimation

Self-supervised depth estimation methods do not rely on real depth labels, but are trained using techniques such as image reconstruction and geometric constraints. Godard et al. [30] proposed Monodepth to improve the continuity of the depth map by ensuring the consistency of left and right disparity. Subsequently, they proposed Monodepth2, introducing a minimum reprojection loss and an automatic masking mechanism to further improve the model's robustness in handling occlusions and dynamic objects [31].

To further improve the precision of depth estimation, Kumar et al. [32] proposed HR-Depth, which utilizes off-the-shelf fine-grained semantic segmentation to better handle complex objects in dynamic scenes. In addition, Kim et al. [33] proposed Local Depth, which improves the accu-

racy of the overall model by training an additional local network to predict the depth gradient and integrating it with the depth prediction from the global network. Building on this foundation, Watson et al. [34] proposed Depth-Hints, which uses depth proxy labels obtained from SGM stereo matching to help the network escape local optima. This approach significantly improves the performance of depth estimation without additional supervision. The methods discussed above enhance the accuracy and robustness of self-supervised monocular depth estimation by incorporating various techniques and architectures. Furthermore, some studies leverage additional semantic supervision to support self-supervised monocular depth estimation. Jung et al. [35] proposed a fine-grained semantic enhancement approach that improves depth estimation by integrating semantic information.

2.3 Advanced Architectures for Depth Estimation

Significant innovations and improvements have been made in network architectures for depth estimation. He et al. [22] introduced the Residual Network (ResNet) architecture, which dramatically enhances training and performance by incorporating residual connections. This approach resolves the common issue of vanishing gradients in deep networks, making deeper network training feasible. Chen et al. [36] applied the inflated convolution technique to improve feature extraction efficiency. By expanding the receptive field of the convolution kernel, this technique captures richer contextual information while maintaining computational efficiency. In recent years, Transformer-based architectures have also been incorporated into depth estimation, further improving accuracy and robustness through powerful sequence modeling capabilities. For instance, DPT [21] provides fine-grained and globally consistent predictions for monocular depth estimation and semantic segmentation tasks by introducing a Transformer. PixelFormer [37] frames depth estimation as a pixel-query refinement problem, improving depth estimation with cross-attention modules and bin-centered predictions. Yang et al. [38] use a Transformer to extract global features and employ an attention gate decoder to capture fine-grained details. Additionally, various constraints have been introduced to enhance algorithms. Patil et al. [39] proposed a method to selectively use coplanar pixel information, enhancing depth estimation. VA Depth [40] employed variational constraints to guide depth prediction, improving accuracy in high-frequency regions by considering first-order variations in the scene.

In monocular depth estimation, Convolutional Neural Networks (CNNs) and Transformers are the dominant approaches, each with certain limitations. CNNs struggle with capturing long-range dependencies, while Transformers, despite excelling in global feature extraction, may overlook local



details. Additionally, the Transformer's attention mechanism, while powerful, is computationally expensive. To address these challenges, this paper proposes a novel multi-scale fusion network architecture, CT-Mono, which integrates local details and global information. In the encoder, multi-scale feature extraction and fusion improve the accuracy of depth estimation, particularly in restoring fine structures in complex scenes. The architecture aims to provide a more efficient solution for depth estimation tasks.

3 Method

3.1 Overview

This paper presents a novel self-supervised monocular depth estimation framework that combines a depth estimation network (DepthNet) and a pose network (PoseNet) to achieve accurate prediction of depth in monocular images. The block diagram of the system is shown in Fig. 2. The system consists of the following two main networks: DepthNet and PoseNet. DepthNet adopts an encoder–decoder structure, where the encoder extracts multi-level features from the input image and then passes these features to the decoder, which processes them to generate depth prediction maps at multiple resolutions. The encoder of DepthNet consists of two types of modules: the Depthwise Separable Dilated Convolution Block (DCB) module and the Enhanced Local–Global Features Fusion (ELGF) module. The PoseNet network, running in parallel with DepthNet, uses a lightweight architecture based on ResNet-18 and is responsible for predicting the camera's pose variations from the input image sequence. To achieve self-supervised learning, the framework converts the depth estimation task into an image reconstruction task. Specifically, the depth map and camera pose predicted are used for image reconstruction, and the network is optimized by minimizing the error between the reconstructed image and the target image. This approach not only enhances the accuracy of depth prediction but also significantly reduces the reliance on annotated data, thereby decreasing the cost of data preparation. The reprojection module plays a crucial role in this process, responsible for mapping points from 3D space back to the 2D image plane to achieve precise image reconstruction. First, the depth map provides the scene depth information corresponding to each pixel, which helps to convert 2D pixel coordinates into points in 3D space. Next, using the camera pose information (including the camera's rotation matrix and translation vector), these 3D points are transformed from the world coordinate system to the camera coordinate system. These 3D points are then mapped back to the image plane according to the camera's projection model, yielding a reconstructed image. This process relies on the camera's internal parameters, that is, the internal structural

parameters (such as focal length and principal point position) that determine how the 3D points are projected onto the 2D image. Ultimately, the reprojection module uses these computations to produce the reconstructed image corresponding to the original image, thus enabling the effective use of the depth map information. The difference between the reconstructed image generated by the reprojection module and the original image can serve as the training loss. By minimizing this loss function, the depth estimation model can progressively reduce the disparity between the reconstructed image and the original image, thus enhancing the prediction accuracy of the depth map.

3.2 DepthNet Encoder

As illustrated in Fig. 3, the depth encoder performs multi-scale feature extraction through the DCB and ELGF modules. The DCB module primarily improves computational efficiency and the ability to capture contextual information by reducing computational complexity and expanding the receptive field, whereas the ELGF module focuses on multi-scale feature extraction and the fusion of local and global features. For a given input image size of $H \times W \times 3$, the image is fed into a convolutional stem and an average pooling layer, respectively. In the convolutional stem, the image is downsampled by a 3×3 convolution, followed by two additional 3×3 convolutions for local feature extraction, after which the feature maps are merged with the input image following average pooling. The merged image is fed into stage 1, where a 3×3 convolution with a sampling step of 2 downsamples the feature map to obtain a feature map of size $\frac{H}{4} \times \frac{W}{4} \times C_1$. Similarly, the output feature map passes through stages 2 and 3 sequentially, and the final output feature map size is $\frac{H}{16} \times \frac{W}{16} \times C_3$. The architecture reduces the spatial dimensionality by combining convolutions to extract local features and average pooling to obtain global information, progressively downsampling the input image. At each stage, a 3×3 convolutional layer is used for feature extraction and dimensionality reduction, and the feature map is progressively reduced to $\frac{1}{4}$, $\frac{1}{8}$, and $\frac{1}{16}$ of the original image. This multi-stage feature extraction process generates richer multi-scale depth information. By fusing the features from different stages, both global and local details in the image are better captured, thus improving the accuracy and robustness of depth estimation.

3.2.1 DCB

As shown in Fig. 4, this study introduces a novel Depthwise Separable Dilated Convolution Block (DCB), designed to efficiently extract multi-scale features while keeping the number of parameters and computational complexity minimal. The DCB module combines the advantages of Depth

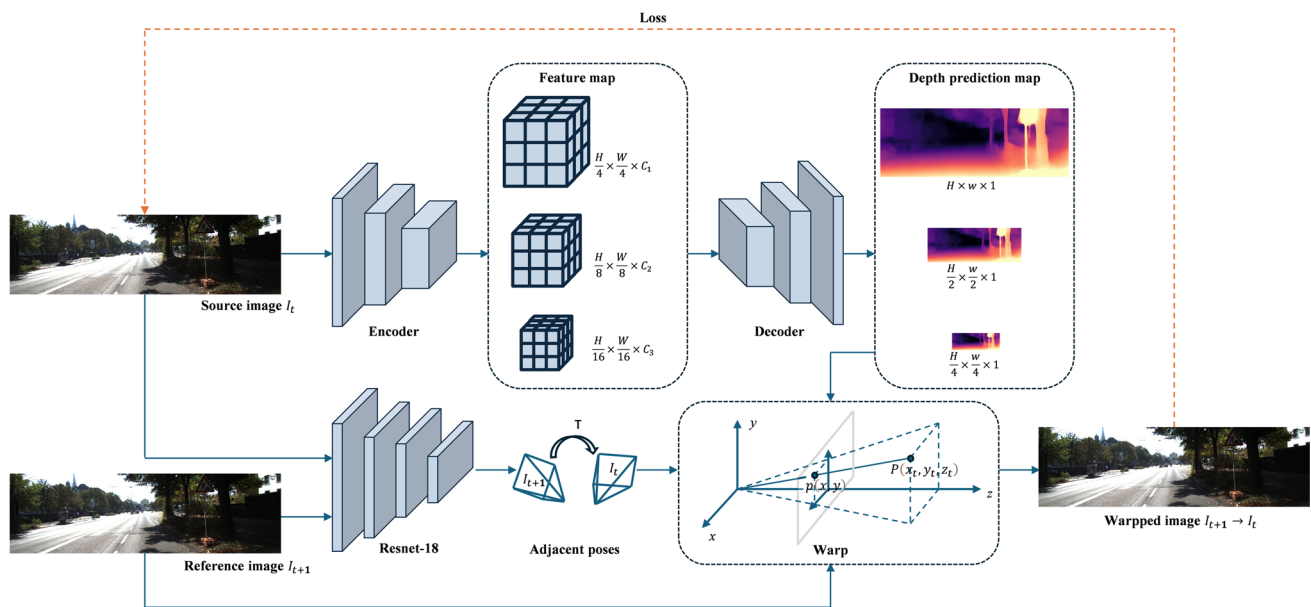


Fig. 2 Overview of the proposed CT-Mono

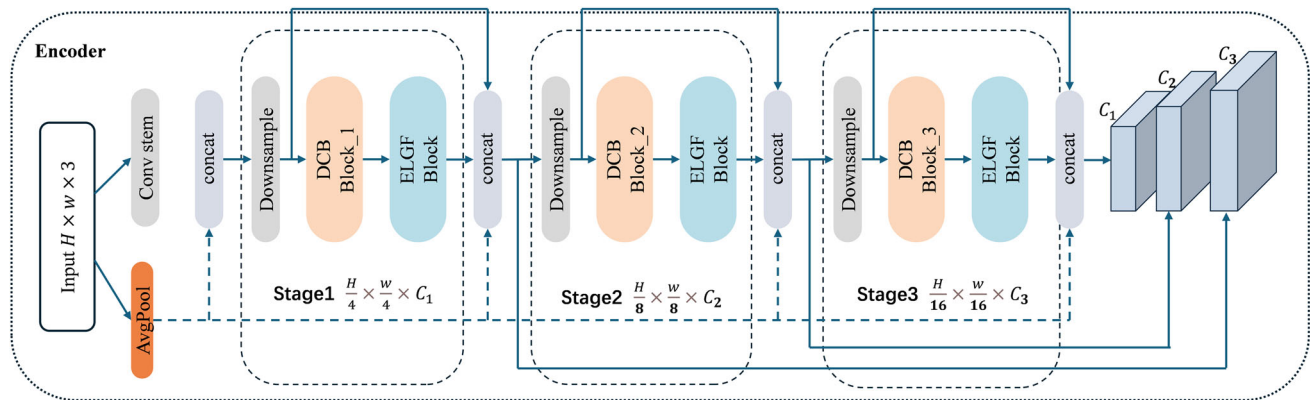


Fig. 3 Depth encoder

Separable Convolution and Cavity Convolution to efficiently capture multi-scale spatial information from the input features by employing different cavitation rates at each layer. Depthwise separable convolution decomposes the traditional convolution operation into two consecutive steps: First, depthwise convolution applies spatial filters independently to each input channel; second, pointwise convolution integrates the output features from each channel. This design significantly reduces the number of parameters and computational costs of the model. In combination with null convolution, the DCB module enhances the capture of contextual information by expanding the effective receptive field of the convolution kernel without increasing the number of parameters.

To enhance the efficiency of feature extraction and the ability to capture multi-scale contextual information, this paper proposes a Depthwise Separable Dilated Convolution Block (DCB), which adopts a parallel multi-branch

design. As illustrated in Fig. 4, the input feature map is first duplicated into several branches. Each branch performs a depthwise convolution with the same kernel size (3×3), but with different dilation rates—such as 1, 2, and 3—corresponding to varying receptive field sizes. These depthwise dilated convolutions are applied independently on each input channel to extract spatial features ranging from local to global scales. Subsequently, the outputs of all branches are concatenated along the channel dimension and fused via a pointwise (1×1) convolution, which also adjusts the number of output channels. This design effectively enhances the model’s multi-scale representation capability while maintaining low parameter count and computational cost.

Compared with conventional depthwise separable convolutions (e.g., those used in MobileNet), the DCB introduces dilated convolutions to further expand the receptive field

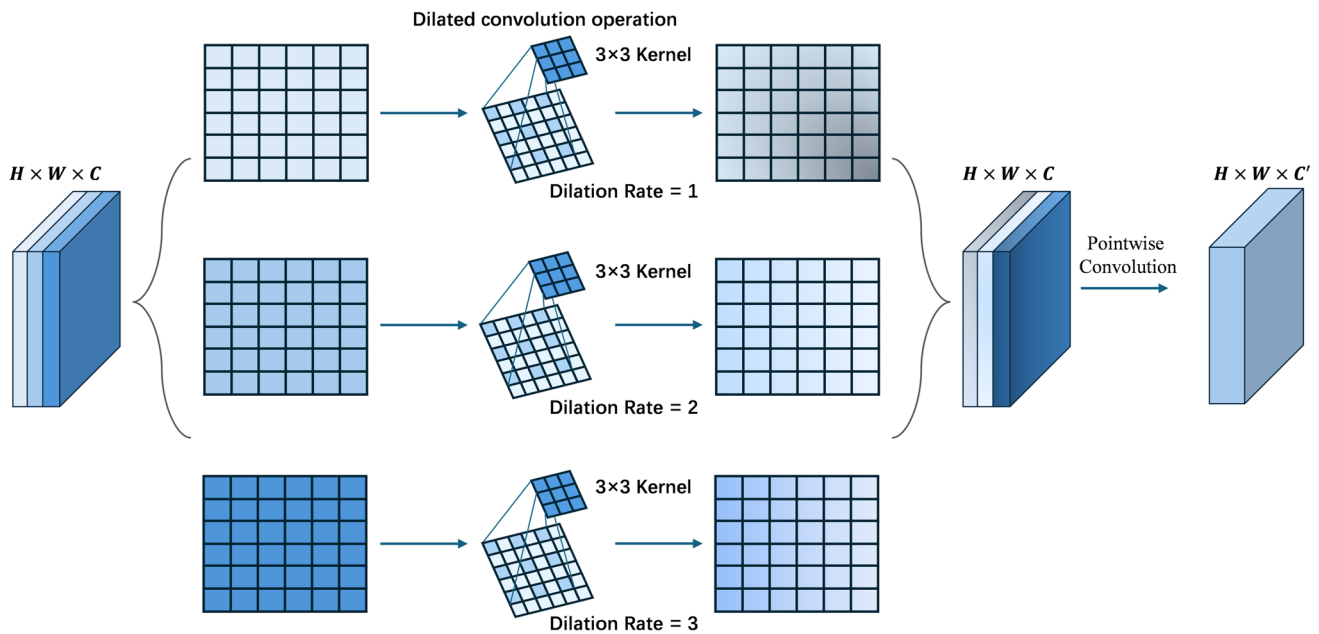


Fig. 4 Depthwise Separable Dilated Convolution Block (DCB)

and improve the modeling of contextual information. While MobileNet emphasizes parameter efficiency, it is limited in capturing long-range dependencies. In contrast, DCB incorporates multiple dilation rates (e.g., 1, 2, 3) in the depthwise convolution stage to extract richer spatial context. Moreover, by cascading convolutional paths with different dilation rates in a parallel manner, the DCB forms a structure capable of more expressive feature extraction in complex scenarios. Therefore, DCB not only retains the lightweight characteristics of depthwise separable convolutions but also significantly enhances multi-scale feature extraction, making it well-suited for vision tasks that demand high representational power.

Conventional depthwise separable convolution consists of two components: depthwise convolution and pointwise convolution. Its ability to significantly reduce the number of parameters and computational load of the model arises from the fact that depthwise convolution handles only the spatial features of a single input channel, while pointwise convolution integrates features across channels. This approach not only improves computational efficiency but also preserves the network's performance. Depthwise convolution applies spatial filters independently to each input channel. For each channel in the input feature map, the output of its depthwise convolution is Y_c :

$$Y_c[i, j] = \sum_{m, n} X_c[i + m, j + n] \cdot K_c[m, n] \quad (1)$$

where X_c denotes the c -th channel of the input feature map, K_c is the convolution kernel for that channel, and m, n denote the indices of the convolution kernel in the vertical and horizontal directions, respectively.

Pointwise convolution is a 1×1 convolution used to integrate the features of each channel obtained by depthwise convolution. For each position, the output of the pointwise convolution is $Z[i, j]$:

$$Z[i, j] = \sum_{c=1}^{C_{\text{out}}} Y_c[i, j] \cdot W_c \quad (2)$$

where W_c is the weight of the 1×1 pointwise convolution kernel at channel c , and C_{out} is the number of output channels.

Null convolution is a technique that extends the standard convolutional kernel by introducing an extension factor called the 'null rate' to increase the receptive field of the kernel, thereby improving the field of view without adding additional computational cost. It is particularly suited for tasks that require capturing a wider range of contextual information. Given an input feature map, its output Y after null convolution is:

$$y[i] = \sum_{k=1}^K x[i + r \cdot k] \cdot w[k] \quad (3)$$

In the formula, $y[i]$ represents the value of the output feature map at position i , $x[i + r \cdot k]$ denotes the input value after calculating the offset based on the null rate r and index k on the input feature map centered at position i , $w[k]$ is the weight of the convolution kernel, and K denotes the size of the convolution kernel.

By fusing traditional depth-separable convolution with cavity convolution, the depth separable cavity convolution module combines the advantages of both techniques, increasing the receptive field by applying cavity convolution in the depthwise convolution step while maintaining parametric efficiency.

$$Y_c[i, j] = \sum_{m, n} X_c[i + r \cdot m, j + r \cdot n] \cdot K_c[m, n] \quad (4)$$

where $Y_c[i, j]$ is the output of the convolution operation corresponding to the value of channel c of the output feature map at position (i, j) , and $X_c[i + r \cdot m, j + r \cdot n]$ denotes an element of the input feature map, where X_c denotes channel c of the input feature map. $[i + r \cdot m, j + r \cdot n]$ computes the actual position of the convolution kernel element, which takes into account the null rate r , causing the convolution kernel to skip $r - 1$ intervals on the input feature map, thereby expanding its receptive field. m and n are the indices of the convolution kernel in the vertical and horizontal directions, respectively, traversing the entire convolution kernel. $K_c[m, n]$ is the weight of the convolution kernel at channel c at position (m, n) , and r is the null rate.

3.2.2 ELGF

The Enhanced Local–Global Features Fusion (ELGF) Module proposed in this paper is primarily divided into two components. The goal of the first component is to combine feature information at different scales by segmenting and processing the input features, thereby enhancing the diversity and richness of the feature representation. This process effectively integrates multiple feature perspectives, thereby enhancing the model's local understanding of the input data. The second component implicitly encodes the global image while maintaining computational efficiency. The first component of our encoder is inspired by EdgeNeXt [41] and utilizes a hierarchical representation for multi-scale processing, which enables a larger spatial receptive field for the output feature representation. The input tensor $H \times W \times C$ is partitioned into 3 subsets. Each subset x_i has the same spatial dimensions, and the number of channels is $\frac{C}{3}$, where $i \in \{1, 2, 3\}$. Except for the first subset, the remaining two subsets undergo one deep 3×3 convolution operation, with the output denoted as y_i . The final output depends on the

summation of the outputs from the 3 subsets, which can be represented by the formula:

$$Y = \sum_{i=1}^3 y_i \quad (5)$$

The feature map Y (with dimensions $H \times W \times C$) is linearly projected onto the query Q , the key K , and the value V :

$$Q = W_Q Y, \quad K = W_K Y, \quad V = W_V Y \quad (6)$$

where W_Q , W_K , and W_V are the weight matrices for the projection, and the outputs are all of size $H \times W \times C$.

To efficiently model global dependencies across channels, we introduce the Cross-Covariance Attention (XCA) mechanism to enhance the input features. Unlike standard self-attention mechanisms that typically operate along the spatial dimensions (i.e., $HW \times HW$), XCA performs attention modeling along the channel dimension ($C \times C$). This results in a computational complexity of $O(C^2)$, which is significantly lower than the $O(H^2W^2)$ complexity of conventional self-attention. Since $C \ll HW$ in most vision tasks, XCA is able to retain global modeling capability while substantially reducing computational overhead. This makes it well-suited not only for high-resolution inputs but also for deployment on resource-constrained edge devices. Moreover, XCA captures the interdependencies and co-variation among feature channels, which helps to improve the discriminability and robustness of feature representations.

Before calculating the cross-covariance attention, L2 normalization of Q and K is required to stabilize the training process. Calculating cross-covariance attention:

$$\text{Attention}(Q, K, V) = V \cdot \text{softmax}(Q^T \cdot K) \quad (7)$$

where $Q^T \cdot K$ has a size dimension of $C \times C$. Since the number of channels is much smaller than the spatial dimensions ($C \ll HW$), applying dot-product operations to the channel dimension can significantly reduce the computational effort. This is especially important for resource-constrained devices (e.g., edge devices), as it reduces computational overhead and memory usage and improves inference speed and real-time performance. Applying cross-covariance attention to augmented inputs X :

$$\tilde{X} = \text{Attention}(Q, K, V) + X \quad (8)$$



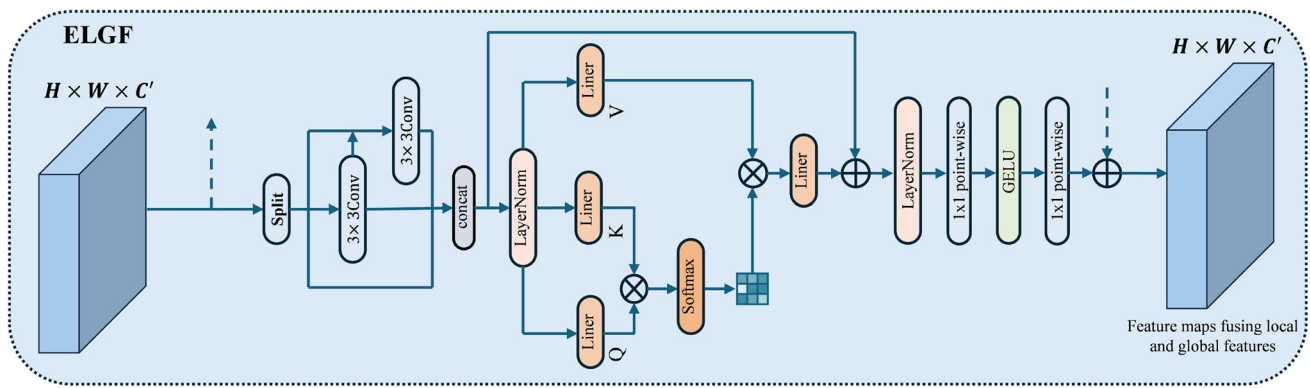


Fig. 5 ELGF (Enhanced Local and Global Features Combination) module

Nonlinear features are then generated using 1×1 point-wise convolution, layer normalization, and the GELU activation function:

$$\hat{X} = X + \text{Linear}_G \left(\text{Linear}(\text{LN}(\tilde{X})) \right) \quad (9)$$

More details about the ELGF module are in Fig. 5.

3.2.3 DepthNet Decoder

The depth decoder is inspired by the decoder module of Rm-sfm [42], which outputs the inverse depth through three upsampling modules, as shown in Fig. 6. In the depth estimation decoder, the network fuses feature maps from multiple scales. The feature maps represent features extracted from different depth levels of the encoder, with varying resolutions and semantic levels, respectively. The resolution of these feature maps is progressively recovered through upsampling and convolution operations, with lower-layer features provided by skip connections being combined at each stage to ensure that the generated depth maps have better resolution and detail preservation. For depth estimation tasks, lower-level features provide rich edge and texture information, while higher-level features contain global information about the scene. The introduction of skip connections allows this information to be fused during the decoding process, resulting in depth maps that contain both global depth trends and local details.

3.3 PoseNet

A pre-trained ResNet-18 is employed as the pose encoder, with the input being a pair of color images. ResNet-18 efficiently extracts image features through its deep structure and residual connections to generate a joint feature representation. These features are then fed into a pose decoder containing four convolutional layers, responsible for estimat-

ing the 6-degree-of-freedom (6-DoF) relative pose between neighboring images. This decoder progressively refines the features through convolutional operations and ultimately outputs the corresponding bit-pose parameters.

3.4 Loss

The task of this paper is to perform self-supervised monocular depth estimation, where, instead of using real depth values as training data, the depth estimation task is converted into an image reconstruction task. This provides a supervised signal under unsupervised conditions by minimizing the image reconstruction loss function. Specifically, the image reconstruction loss consists of the following components:

1. **Photometric Reprojection Loss [43]:** Reprojection loss is used to assess the difference between a synthetic image generated from depth information and the original image. The process typically involves reprojecting the depth map into the camera's view to generate a synthetic image. By minimizing the difference between the synthetic image and the target image, depth estimation is optimized, thereby improving the model's understanding of the scene. The formula is:

$$L_p(\hat{I}_t, I_t) = \alpha \frac{(1 - \text{SSIM}(\hat{I}_t, I_t))}{2} + (1 - \alpha) \|\hat{I}_t - I_t\| \quad (10)$$

$$\hat{I}_t = F(I_s, P, D_t, K) \quad (11)$$

where $\hat{I}_t$ is the synthesized target image, obtained from the source image I_s , estimated pose P , predicted depth D_t , and camera internal reference K . I_t is the real target image. In Eq. (10), SSIM (Structural Similarity Index) measures the structural similarity between the reconstructed image and the target image, while L1 loss measures the pixel-level difference between them. α is a weighting factor used to balance these two, and it is set to 0.85.

2. **Minimum Photometric Loss [31]:** Minimum photometric loss optimizes depth estimation and image reconstruction

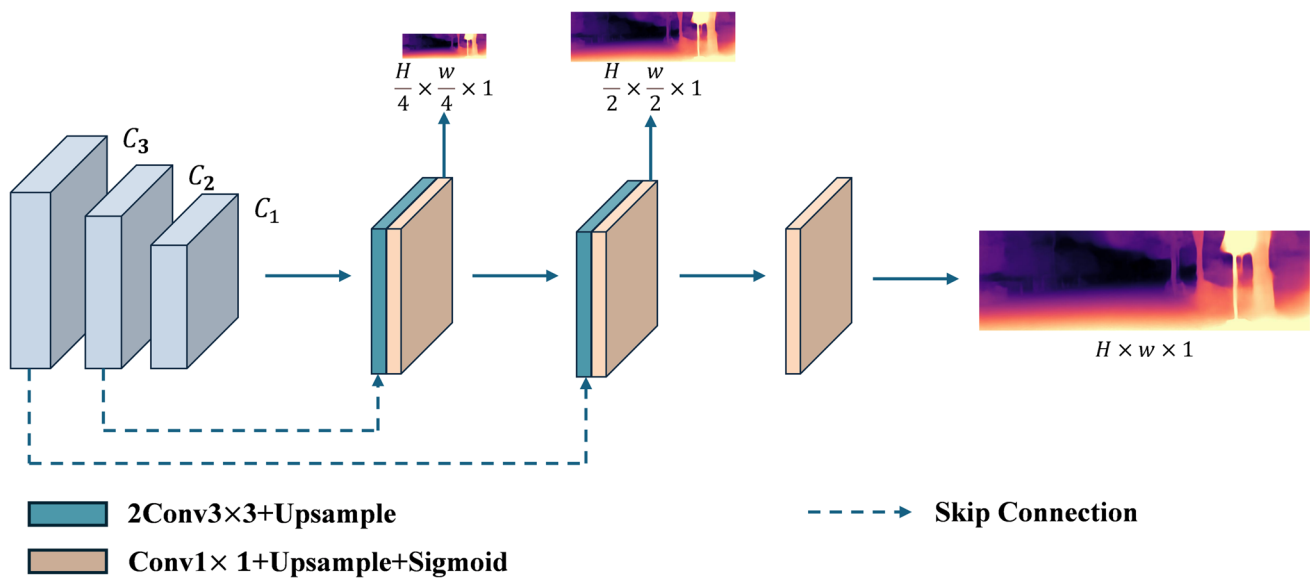


Fig. 6 DepthNet decoder

by comparing pixel values between multiple frames to find the smallest photometric difference. To handle pixels outside the field of view and occlusions in the source image, the minimum photometric loss calculates the loss between multiple frames and selects the minimum value. The model can effectively deal with occlusions and pixels outside the field of view because it does not rely on information from a single viewpoint. The formula is:

$$L_p(I_s, I_t) = \min_{I_s \in \{-1, 1\}} L_p(\hat{I}_t, I_t) \quad (12)$$

wherein I_s may be a previous frame or a later frame of the target image.

3. Binary Mask [44]: In self-supervised monocular depth estimation, the model relies on image information between successive frames to estimate depth and camera motion. However, moving objects in the scene can lead to incorrect depth estimates, as they violate the parallax assumption. The parallax assumption is based on camera motion in a stationary scene, whereas moving objects introduce complex motion patterns that lead to errors. Therefore, the use of a binary mask to remove these moving pixels helps improve the robustness and accuracy of the model. A mask value of 1 indicates that the reprojection error of the pixel is minimized, and a value of 0 indicates that the pixel may belong to a moving object and should be excluded from the loss calculation.

$$\mu = \min_{I_s \in \{-1, 1\}} L_p(I_s, I_t) > \min_{I_s \in \{-1, 1\}} L_p(\hat{I}_t, I_t) \quad (13)$$

The final image reconstruction loss is adjusted using the generated binary mask, with only the loss of static pixels being calculated:

$$L_r(\hat{I}_t, I_t) = \mu \cdot L_p^{\min}(I_s, I_t) \quad (14)$$

To ensure that the model can accurately reconstruct the target image while retaining important edge features, an edge-aware smoothing loss is introduced to enhance the smoothness and consistency of the depth map:

$$L_{\text{smooth}} = \sum_{i,j} \left(\left| \partial_x d_{ij}^* \right| e^{-|\partial_x I_{ij}|} + \left| \partial_y d_{ij}^* \right| e^{-|\partial_y I_{ij}|} \right) \quad (15)$$

where $\partial_x d_{ij}^*$ and $\partial_y d_{ij}^*$ denote the gradient of the generated depth map d_{ij} in the x-direction and y-direction, and $\partial_x I_{ij}$ and $\partial_y I_{ij}$ denote the gradient of the image I_{ij} in the x-direction and y-direction.

$$d_{ij}^* = \frac{d_{ij}}{\bar{d}} \quad (16)$$

where d_{ij}^* is the mean-normalized inverse depth, and $\bar{d}$ denotes the mean of the depth map.

In addition to the aforementioned loss terms, we also explored the gradient consistency loss, which aims to enhance edge structure preservation by enforcing consistency between the gradient directions of the predicted depth map and the input image. This type of loss can be formalized



as a discrepancy measure in the gradient domain between the image and the depth map:

$$L_{\text{grad}} = \sum_{i,j} |\nabla d_{i,j} - \nabla I_{i,j}| \quad (17)$$

We observed that this loss exhibits a strong ability to preserve structural details in edge regions. However, it tends to be sensitive to gradient perturbations during the early stages of training, and its overall contribution to performance improvement is limited. Considering that the edge-aware smoothness loss offers a better balance between enforcing smoothness and preserving edges, we ultimately adopt it as the main regularization term in our framework.

4. The final total loss function is the weighted sum of all the above loss functions and is given by the following formula:

$$L_{\text{total}} = L_r(\hat{I}_t, I_t) + \lambda_{\text{smooth}} L_{\text{smooth}} \quad (18)$$

where λ_{smooth} and λ_{RRCE} are weight parameters to adjust the effects of smoothing loss and RRCE loss.

4 Experiments

This section evaluates the proposed framework, demonstrates the performance of CT-Mono and compares it with other representative methods.

4.1 Datasets

KITTI was used as our test benchmark [45]. It contains 61 three-dimensional road scenes for autonomous driving and robotics research, collected by several sensors, including cameras, 3D LIDAR, GPU/IMU, etc. To train and evaluate the proposed method, KITTI Eigen segmentation was used to evaluate deep prediction. In total, 39,180 monocular triples were used for training, 4,424 for evaluation and 697 for testing. To ensure fairness, the depth of the predictions was limited to the range [0, 80]m. For more details on the evaluation, please refer to [25].

4.2 Implementation Details

In this study, we implemented our proposed method using the PyTorch framework and trained it on a computer equipped with NVIDIA GeForce RTX 4090 GPUs. The batch size was set to 12, with a total of 35 training epochs. The AdamW optimizer [42] was used, with a weight decay parameter of 1×10^{-2} and an initial learning rate of 1×10^{-4} , requiring roughly 20h to train. To assess model performance, we used

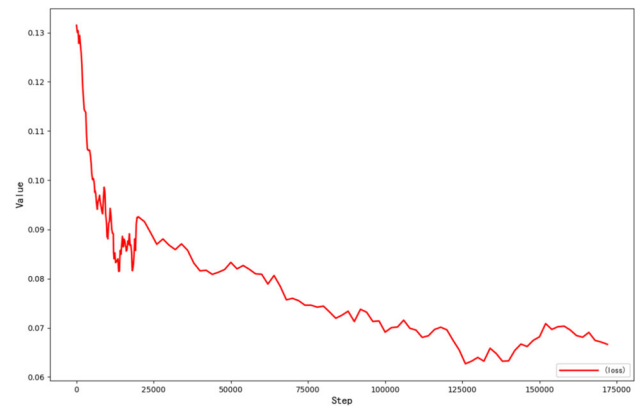


Fig. 7 Decreasing trend of training loss with training steps

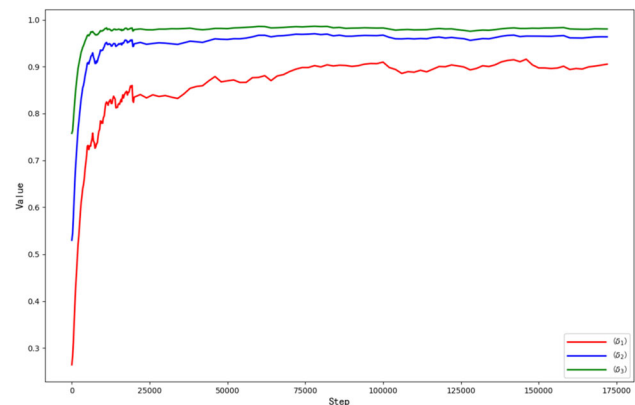


Fig. 8 Change in scale invariance accuracy metrics with training step

seven standard metrics commonly applied in self-supervised monocular depth estimation (MDE) model comparisons: absolute relative error (AbsRel), squared relative error (Sq Rel), root mean square error (RMSE), root mean square error in logarithmic terms (RMSElog), and three scale-invariant accuracy metrics $\delta_1 < 1.25$, $\delta_2 < 1.25^2$, and $\delta_3 < 1.25^3$. These metrics provide a comprehensive evaluation of the model's accuracy and robustness in depth prediction tasks.

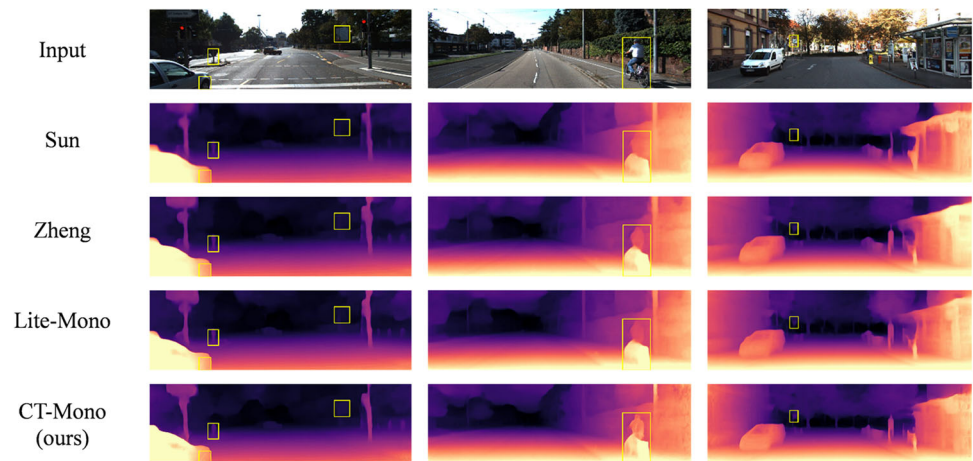
4.3 Training Results

The model's performance, including training loss and accuracy, was recorded at the end of each training epoch. In the initial training epochs, the rapid decrease in the model's loss function generally indicates that the model begins to learn useful features from the data. As shown in Fig. 7, the dynamics of the model's loss function throughout training are illustrated. A significant decrease in the loss function is observed in the initial phase, from the beginning of training until approximately 25,000 steps, indicating the model's ability to quickly extract useful information from the data and begin optimization. The rapid reduction in a loss at this stage can be attributed to the rapid adjustment of model param-

Table 1 Compares CT-Mono with previous models on the KITTI Eigen split

Method	Architecture	W × H	AbsRel ↓	Sq Rel ↓	RMSE ↓	RMSElog ↓	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Monodepth2 no pt [31]	CNN	640 × 192	0.130	1.144	5.485	0.232	0.831	0.932	0.968
Monodepth2 S [31]	CNN	640 × 192	0.109	0.873	4.960	0.209	0.864	0.948	0.975
FreqAwareDepth [36]	CNN	640 × 192	0.112	0.834	4.746	0.189	0.880	0.961	0.982
WaveletMonodepth [46]	CNN	640 × 192	0.109	0.845	4.800	0.196	0.870	0.956	0.980
Depth-Hints [47]	CNN	640 × 192	0.109	0.845	4.800	0.196	0.870	0.956	0.980
MT-SfMLearner [48]	Transformer	640 × 192	0.112	0.838	4.771	0.188	0.879	0.960	0.982
VTDepthB0 [49]	Transformer	640 × 192	0.113	0.854	4.797	0.190	0.879	0.960	0.982
VTDepthB1 [49]	Transformer	640 × 192	0.109	0.797	4.598	0.186	0.887	0.963	0.982
HRFormer [50]	Transformer	640 × 192	0.108	0.812	4.634	0.185	0.891	0.963	0.982
DPT* [51]	Transformer	640 × 192	0.210	0.847	4.554	0.184	0.847	0.948	0.976
CADepth [52]	CNN+Transformer	640 × 192	0.107	0.849	4.885	0.204	0.869	0.951	0.976
Zheng [53]	CNN+Transformer	640 × 192	0.111	0.905	4.749	0.189	0.886	0.961	0.981
Lite-Mono [44]	CNN+Transformer	640 × 192	0.107	0.765	4.561	0.183	0.886	0.963	0.983
CT-Mono (ours)	CNN+Transformer	640 × 192	0.108	0.801	4.611	0.185	0.883	0.961	0.982
Monodepth2 [31]	CNN	1024 × 320	0.115	0.882	4.701	0.190	0.879	0.961	0.982
Sun [54]	CNN	1024 × 320	0.110	0.791	4.557	0.184	0.887	0.964	0.983
MT-SfMLearner [54]	Transformer	1024 × 320	0.104	0.799	4.547	0.181	0.893	0.963	0.982
Zheng [53]	CNN+Transformer	1024 × 320	0.105	0.826	4.582	0.182	0.896	0.965	0.982
Song [55]	CNN+Transformer	1024 × 320	0.111	0.779	4.574	0.184	0.882	0.964	0.984
Lite-Mono [40]	CNN+Transformer	1024 × 320	0.102	0.746	4.444	0.179	0.896	0.965	0.983
CT-Mono (ours)	CNN+Transformer	1024 × 320	0.102	0.738	4.430	0.179	0.896	0.965	0.982

The W × H columns indicate the image size of the input models. The best results are highlighted using bolding. The metrics are error metric ↓ and accuracy metric ↑

Fig. 9 Visualization comparison results between CT-Mono and other models on the KITTI dataset at an input resolution of 1024 × 320

Table 2 Comparison of computational efficiency of different models during training and inference phases

Method	FPS	GPU memory usage (GB)	Para (M)	Speed (ms)
Monodepth2	71.2	9.73	14.84	41.3
CADepth	49.7	20.12	18.74	128
Lite-Mono	81.6	10.18	3.1	47.6
CT-Mono (ours)	77.4	9.88	3.4	48.2



ters from a random initialization, which reduces prediction error. From around 50,000 steps until the end of training, the loss curve shows a smooth downward trend, suggesting that the model parameters are adjusted more steadily, leading to model stabilization. This gradual optimization suggests effective model convergence, with the loss function's continued decrease reflecting the model's adaptation to the training data and improvement in generalization ability.

Figure 8 illustrates the variation in model accuracy under different scale-invariance conditions. The scale-invariance accuracy metrics $\delta_1 < 1.25$, $\delta_2 < 1.25^2$, $\delta_3 < 1.25^3$ measure the model's prediction accuracy across scales. As shown in the figure, all accuracy metrics increase rapidly from the beginning of training until approximately 25,000 steps, indicating that the model quickly learns and adapts to the task of scale prediction during this stage. These accuracy metrics then enter a stabilization phase. This initial rapid improvement likely reflects the model's progression from poor initial predictions to a higher prediction level through initial adaptation. This rapid learning may result from efficient gradient propagation and a large learning rate, enabling the model to learn quickly from substantial data. The onset of this stabilization phase suggests that the model is approaching the upper limit of its predictive capacity, shifting the focus to fine-tuning and optimization.

4.4 Comparative Results

To validate the effectiveness of the proposed CNN-Transformer hybrid architecture, we conducted comparisons with representative CNN-based, Transformer-based, and hybrid models under the same input resolutions (640×192 and 1024×320). The results are summarized in Table 1. At a resolution of 640×192 , our method CT-Mono (Ours) achieved an AbsRel of 0.108 and a Sq Rel of 0.801, outperforming most pure CNN-based methods (e.g., Monodepth2-S with AbsRel = 0.109, Sq Rel = 0.873) and pure Transformer-based methods (e.g., VTDepthB0 with AbsRel = 0.113, and HRFormer with AbsRel = 0.108). Although HRFormer matches our method in AbsRel, it exhibits a higher Sq Rel, indicating a less favorable error distribution. When further evaluated at a higher resolution of 1024×320 , CT-Mono achieves an AbsRel of 0.102 and a Sq Rel of 0.738, representing the best performance among all compared methods. In contrast, the most competitive Transformer-based method, MT-SfMLearner, reports an AbsRel of 0.104 and a Sq Rel of 0.799, while representative CNN-based methods, such as that of Sun et al., report AbsRel = 0.110 and Sq Rel = 0.791—both slightly inferior to our approach. These results clearly demonstrate that the hybrid architecture combining the local modeling capabilities of CNNs with the global representation strength of Transformers enables superior accuracy and better error control. Moreover, compared to existing CNN-

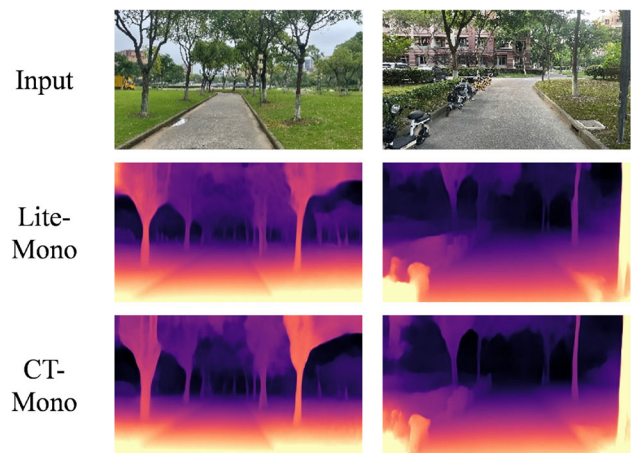
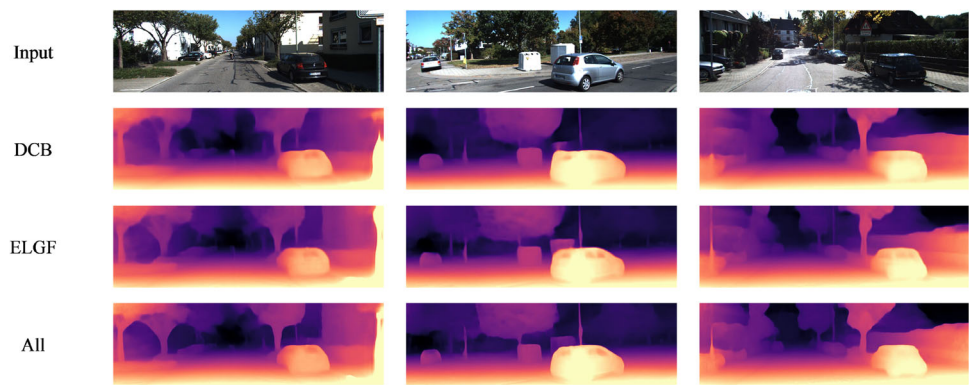


Fig. 10 Comparison of depth map results between CT-Mono and Lite-Mono for realistic real pictures at an input resolution of 1024×320

Transformer hybrid methods (e.g., Lite-Mono, CAdDepth, and Zheng et al.), CT-Mono exhibits competitive or even superior performance across most metrics, particularly in RMSE and RMSElog, where it shows enhanced stability. This further confirms the effectiveness of our hybrid design strategy.

We observed that under the same conditions, our model is able to capture and represent depth information in complex scenes more accurately. Specifically, our model demonstrates superior performance in extracting fine-grained image details and achieving high depth estimation accuracy, as illustrated in Fig. 9. From the visual results, it is evident that the CT-Mono model achieves better accuracy and continuity in depth prediction. Furthermore, CT-Mono shows clear advantages in handling edges and object boundaries. For instance, in the middle column of the figure, the boundaries between the bicycle and the sidewalk are distinctly preserved in the depth map predicted by CT-Mono, whereas other models exhibit inferior performance in these boundary regions. Overall, the CT-Mono model demonstrates significant superiority in detailed depth estimation, edge preservation, and overall depth map continuity.

In Table 2, we compare the computational efficiency of CT-Mono with other mainstream monocular depth estimation methods. It is important to note that the FPS and GPU memory usage reported in the table are measured during the training phase: FPS indicates the number of image frames processed per second during training, reflecting training efficiency; GPU memory refers to the amount of video memory occupied during training. The Speed (ms) metric corresponds to the inference phase, representing the average time (in milliseconds) required to process a single image during testing or deployment, which directly relates to the model's real-world responsiveness.

Fig. 11 Comparison of visualization results of ablation experiments**Table 3** Comparison of generalization performance of different models on the Make3D dataset

Method	AbsRel	Sq Rel	RMSE	RMSE log
Monodepth2	0.322	3.589	7.417	0.163
Zhou	0.383	5.321	10.470	0.478
CADepth	0.312	3.086	7.066	0.159
Lite-Mono	0.305	3.060	6.981	0.158
CT-Mono (ours)	0.307	3.052	6.979	0.159

Bold values indicate the best performance among all methods

From the table, it can be seen that CT-Mono has only 3.4 million parameters, close to Lite-Mono's 3.1 million, and significantly fewer than Monodepth2 and CADepth, demonstrating strong model compression capability. Regarding training efficiency, CT-Mono achieves 77.4 FPS, nearly matching Lite-Mono and outperforming Monodepth2 (71.2 FPS) and CADepth (49.7 FPS), while maintaining relatively low GPU memory consumption (9.88 GB). Although the single-frame inference time of CT-Mono is 48.2 ms, slightly higher than Monodepth2's 41.3 ms, it achieves a good balance between high accuracy and operational efficiency.

In addition to the comparison tests on the laboratory dataset, we evaluated the CT-Mono model using real-world images captured with an iPhone 12. The model's practical applicability was further validated through depth estimation on real scenes. As shown in Fig. 10, the CT-Mono model continued to demonstrate superior performance in real-image tests. These tests demonstrate that the CT-Mono model not only performs well in a laboratory environment but also shows high reliability and accuracy in real applications, making it suitable for depth estimation tasks across diverse real-world scenarios.

To further validate the model's generalization capability beyond structured datasets such as KITTI, we conducted cross-dataset evaluation on the Make3D dataset. It is important to emphasize that no training or fine-tuning was performed on Make3D; the model was trained solely on the KITTI dataset. The training process used the same hardware

and parameter settings, specifically an NVIDIA RTX 4090D GPU, a batch size of 12, and an input resolution of 640×192 .

As shown in Table 3, despite the lack of training or fine-tuning on Make3D, CT-Mono demonstrates stable performance on this dataset. Its absolute relative error (AbsRel) is 0.307, slightly higher than Lite-Mono's 0.305, yet significantly better than Monodepth2 and Zhou's methods, indicating relatively accurate average error control. The squared relative error (Sq Rel), which emphasizes large errors, shows CT-Mono's advantage with the lowest value of 3.052, reflecting its effectiveness in mitigating extreme prediction errors. The root mean squared error (RMSE) and logarithmic RMSE (RMSE log) are 6.979 and 0.159, respectively, nearly matching Lite-Mono and outperforming other compared methods, demonstrating strong performance in both overall magnitude and proportional error. These results indicate that CT-Mono possesses robust cross-dataset generalization ability, effectively transferring from the structured KITTI scenarios to the more diverse and complex natural scenes in Make3D, thus exhibiting strong robustness and practical applicability.

4.5 Ablation Experiments

In this study, to verify the specific contributions of the proposed modules to the model's performance, we conducted ablation experiments, with results summarized in Table 4. First, the baseline model containing only the DCB module exhibited relatively inferior performance on error metrics such as AbsRel, Sq Rel, and RMSE, as well as lower accuracy scores $\delta_1 < 1.25$, $\delta_2 < 1.25^2$, $\delta_3 < 1.25^3$, indicating that relying solely on DCB for depth information modeling has certain limitations.

Subsequently, introducing the ELGF module without the local modeling branch (denoted as ELGF w/o Local) on top of DCB led to a significant improvement. Specifically, AbsRel decreased from 0.119 to 0.111, RMSE dropped correspondingly to 4.640, and accuracy metrics simultaneously increased, demonstrating that the global guidance branch effectively enhances feature representation capability. Sim-



Table 4 Ablation study of model architectures

Architecture					Error Metric ↓				Accuracy Metric ↑		
DCB	ELGF w/o Local	ELGF w/o Attn	ELGF	Full ELGF+DCB	AbsRel	Sq Rel	RMSE	RMSElog	δ_1	δ_2	δ_3
✓					0.119	1.144	5.485	0.232	0.831	0.932	0.968
✓	✓				0.111	0.815	4.640	0.186	0.880	0.961	0.981
✓		✓			0.116	0.835	4.680	0.188	0.878	0.960	0.981
			✓		0.110	0.827	4.678	0.188	0.880	0.958	0.981
✓				✓	0.108	0.801	4.611	0.185	0.883	0.961	0.982

All models were trained and tested on KITTI with an input size of 640×192

The best results are highlighted using bolding. The metrics are error metric ↓ and accuracy metric ↑

ilarly, the ELGF variant without the attention mechanism (ELGF w/o Attn) also outperformed the standalone DCB, although the improvement was slightly less pronounced than when the local modeling branch was removed. These comparisons highlight that the local modeling component within ELGF plays a critical role in enhancing fine-grained perception, while the attention mechanism further strengthens inter-feature interactions and improves semantic consistency. Furthermore, to analyze the relationship between DCB and ELGF, we evaluated a model retaining the full ELGF module but without DCB. The results showed that this configuration achieved performance on AbsRel and RMSElog close to that of the model with DCB included, though with slight declines in δ_2 and δ_3 . This suggests that ELGF, by incorporating global context modeling combined with local refinement, possesses strong feature representation capability, while DCB remains valuable for supplementing low-level multi-scale structural information. Ultimately, when DCB is combined synergistically with the full ELGF structure, the model attains optimal performance: AbsRel decreases to 0.108, Sq Rel and RMSE reduce to 0.801 and 4.611, respectively, and RMSElog reaches 0.185. Meanwhile, accuracy metrics δ_1 , δ_2 , and δ_3 all achieve their highest values of 0.883, 0.961, and 0.982, respectively.

Although DCB provides a certain degree of multi-scale feature information, its design primarily focuses on static, parallel modeling across scales. In contrast, ELGF introduces an attention-based global guidance structure that, when combined with local modeling, enables dynamic fusion across spatial dimensions and strengthens feature responses in key regions. The experimental results clearly demonstrate that these two modules are not redundant but rather complementary in feature modeling from different perspectives. DCB establishes a stable multi-scale perception foundation, while ELGF further enhances global context awareness and local response selection. Their joint use improves the depth estimation model's accuracy and robustness in modeling scene structures.

Figure 11 shows the visualization results of the ablation experiment. The first row shows the input image. The SEC-

ond row shows the depth estimation results using only the DCB method. The method performs better for depth estimation of close objects, capturing the details and shapes of close objects more accurately, and showing high accuracy especially in depth changes of objects such as vehicles and buildings on the street. However, the DCB method shows significant distortion when dealing with distant objects. The third row shows the results after using the ELGF method, which is able to retain the depth information of distant objects better than the DCB method, especially on distant objects (e.g., distant buildings or roads) in urban backgrounds, where the transition of the depth maps is smoother and the details are better preserved. However, the ELGF method has some shortcomings when dealing with the edges of objects in complex scenes, especially in the more blurred or occluded regions, where the depth estimation shows some blurring and the edge contours are not sharp enough, leading to a certain degree of error. The fourth line shows the complete model output combining the DCB and ELGF methods. By combining the advantages of DCB and ELGF, the full model achieves a better balance between depth estimation accuracy and detail retention. The method is able to maintain the high accuracy of the DCB method when dealing with close objects, while retaining more details in distant and complex backgrounds like ELGF.

5 Conclusion

In this paper, we propose a novel self-supervised monocular depth estimation framework, CT-Mono, which integrates Convolutional Neural Networks (CNNs) with a Transformer architecture. The method introduces a Depthwise Separable Dilated Convolution Block (DCB) to effectively enlarge the model's receptive field, thereby improving depth estimation accuracy. The Enhanced Local and Global Features Combination module (ELGF) combines parallel convolutions with attention mechanisms to achieve a balanced trade-off between detail preservation and contextual awareness, enabling CT-Mono to more precisely reconstruct structural

information in complex scenes. Experimental results on the KITTI dataset demonstrate that CT-Mono outperforms existing methods across multiple evaluation metrics, and visualization on real images further validates its practical applicability. Despite its strong performance in most scenarios, CT-Mono may still exhibit inaccurate estimations on reflective surfaces, low-light conditions, or textureless regions, primarily due to the lack of reliable structural information for reconstruction. Future work will focus on enhancing the model's robustness in challenging environments, such as incorporating illumination-invariant feature extraction, semantic-guided depth constraints, or leveraging multimodal data fusion to improve generalization capabilities.

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Declarations

Conflict of interest The authors declare no Conflict of interest.

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